HOSPITAL EMERGENCY

DASHBOARD

This project analyzes emergency room patient data to monitor hospital performance and improve operational efficiency. The dashboard helps stakeholders track patient visits, waiting time, satisfaction levels, and admission trends.

OBJECTIVE

The objective of this project is to create an interactive dashboard to analyze hospital emergency room data and support better decision-making.

Main goals:

- ❑ Monitor patient admission trends
- ❑ Track average waiting time
- ❑ Analyze patient demographics
- ❑ Evaluate department referrals
- ❑ Provide detailed patient-level information
- ❑ Improve hospital service efficiency

TOOL USED

Microsoft Excel

- ❑ Data storage
- ❑ Data preparation
- ❑ Initial dataset creation

Power BI

- ❑ Data transformation (Power Query)
- ❑ Data modeling
- ❑ DAX calculations
- ❑ Dashboard visualization

STEPS IN PROJECT

- ✓ Requirement Gathering/ Business Requirements
- ✓ Data Walkthrough
- ✓ Data Connection
- ✓ Data Cleaning / Quality Check
- ✓ Data Modeling
- ✓ Data Processing
- ✓ DAX Calculations
- ✓ Dashboard Lay outing
- ✓ Charts Development and Formatting
- ✓ Dashboard / Report Development
- ✓ Insights Generation

Power Query Tranformation Steps

In Power BI Power Query, the following transformations were applied

- ✓ **Source :** Connected Excel dataset to Power BI.
- ✓ **Promoted Headers :** Converted the first row into column headers.
- ✓ **Changed Data Type :** Corrected column data types such as date, number, and text.
- ✓ **Renamed Columns :** Standardized column names for better understanding.
- ✓ **Replaced Values :** Fixed inconsistent values such as gender abbreviations or missing data.
- ✓ **Added Custom Column :** Created calculated fields such as Age Group or Time Category.
- ✓ **Removed Columns :** Deleted unnecessary columns not required for analysis.

KPI'S Requirements

To enhance operational efficiency and provide actionable insights into emergency room performance, we need to create a Hospital Emergency Room Analysis Dashboard in Power BI. This solution will enable stakeholders to track, analyze, and make data-driven decisions regarding patient management and service optimization.

- **KPI 1: Number of Patients**

Measure the total number of patients visiting the ER daily.

Display a daily trend using an area sparkline to understand patterns over time, such as peak days or seasonal trends.

- **KPI 2: Average Wait Time**

Calculate the average time patients wait before being attended to by a medical professional.

Use an area sparkline to show daily fluctuations and identify days with higher wait times that may require operational adjustments.

- **KPI 3: Patient Satisfaction Score**

Analyze the average satisfaction score of patients on a daily basis to evaluate the quality of service provided.

Present a daily trend using an area sparkline to identify dips in satisfaction and correlate them with operational challenges or peak times.

- **KPI 4: Number of Patients Referred**:

Count the number of patients referred to specific departments from the ER each day.

Use an area sparkline to track daily trends and identify departments with high referral rates, which may require additional resources.

Dashboard 1: Monthly View

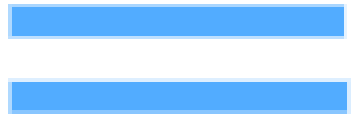
Objective: **Monitor key metrics and trends on a month-by-month basis to identify patterns and areas for improvement.**

❖ Patient Admission Status

Out of the total patients:
2157 patients (49.72%) were admitted
2181 patients (50.28%) were not admitted
Admission status helps identify the **severity of cases handled by the emergency department**.

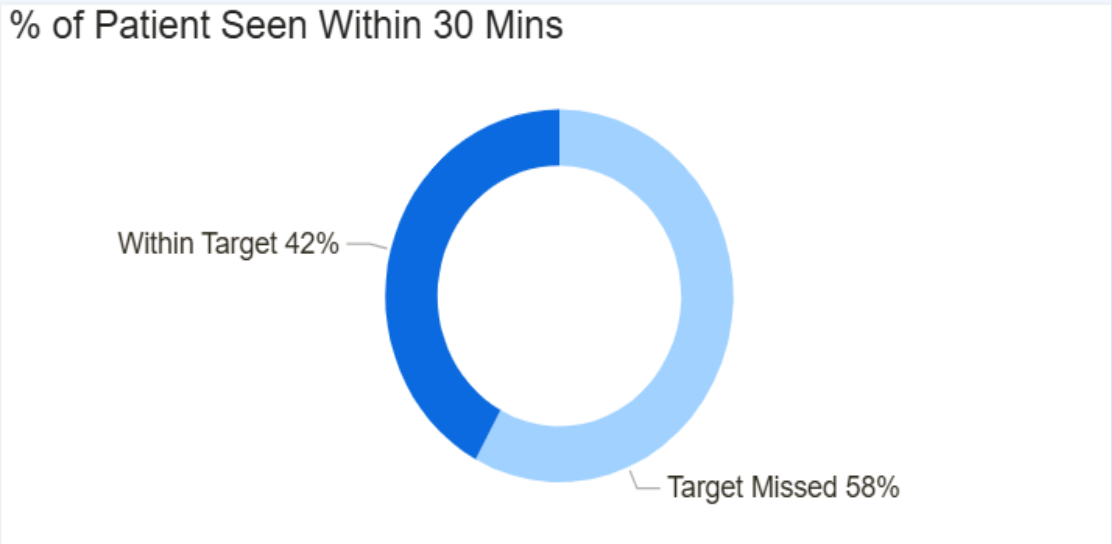
Patients Admission Status

Admission Status	Patients	% of Total
Admitted	2157	49.72%
Not Admitted	2181	50.28%



❖ Patients Seen Within 30 Minutes

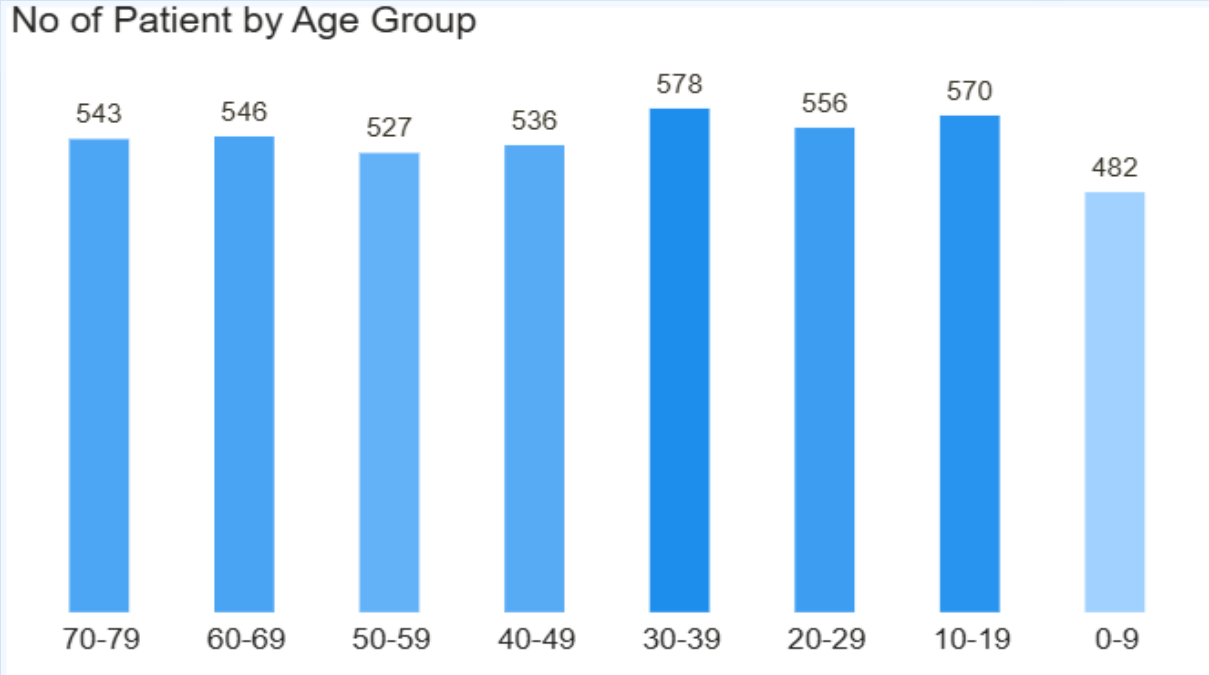
Hospital service performance shows that:
42% of patients received medical attention within 30 minutes
58% of patients experienced longer waiting times
Monitoring this metric helps hospitals evaluate **response efficiency and identify opportunities for service improvement**.



Dashboard 1: Monthly View

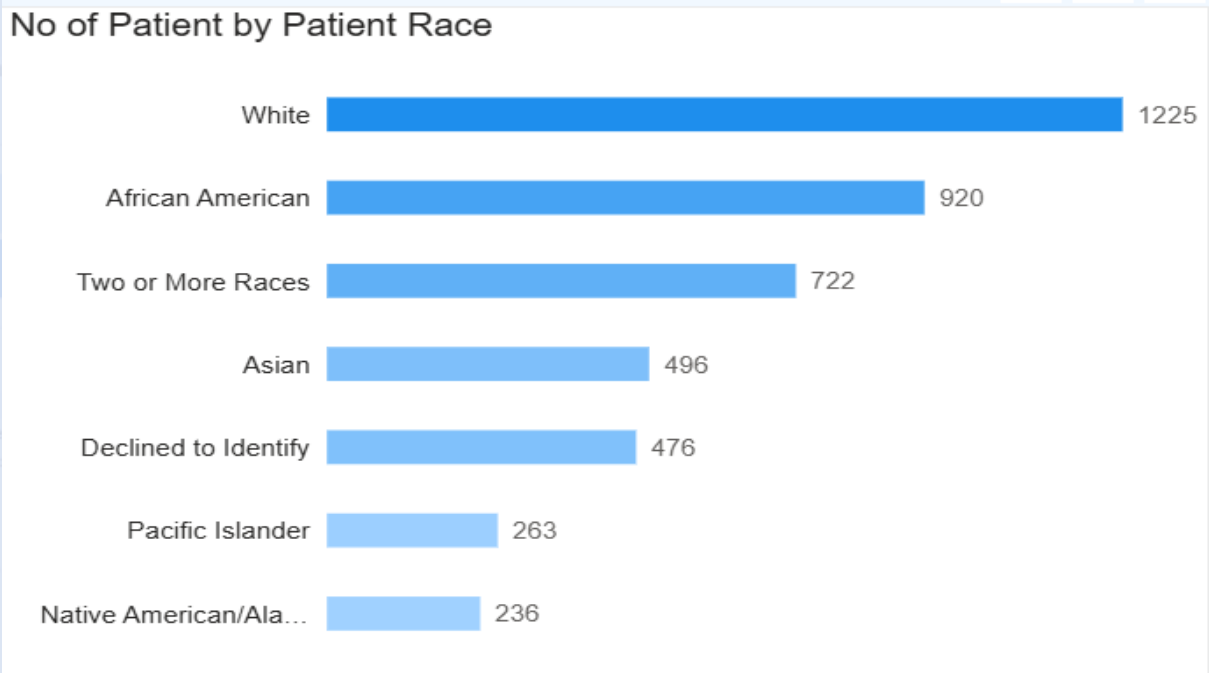
❖ Patient Age Group Distribution

Patient visits are distributed across different age groups. The **30–39 age group records the highest number of visits**, followed by **10–19 and 20–29 age groups**, while the **0–9 age group records the lowest patient count**.
Understanding age distribution helps hospitals plan **age-specific healthcare services and resources**.



❖ Patient Race Distribution

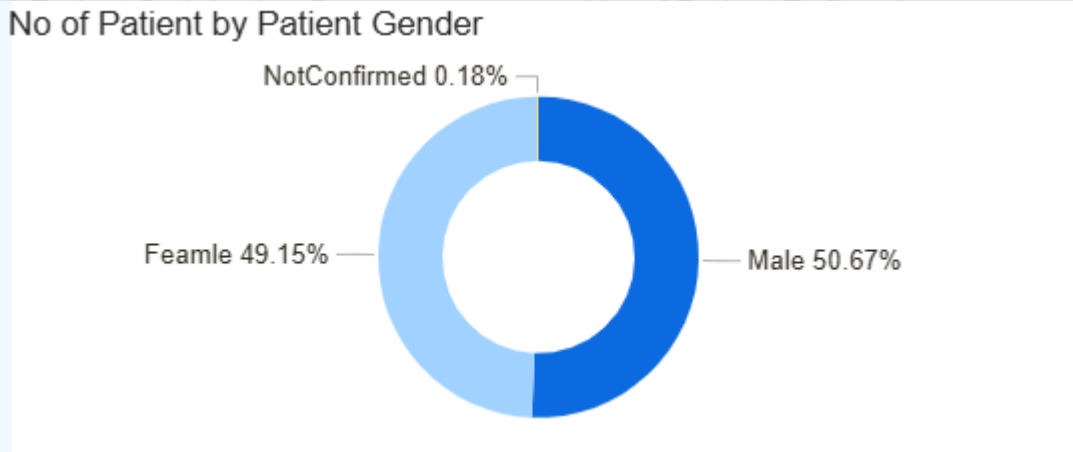
The patient population represents diverse demographic groups. **White patients represent the largest proportion**, followed by **African American and individuals identified as Two or More Races**. Other groups include **Asian, Pacific Islander, Native American, and Declined to Identify**.
Demographic analysis supports **inclusive healthcare planning and resource allocation**.



Dashboard 1: Monthly View

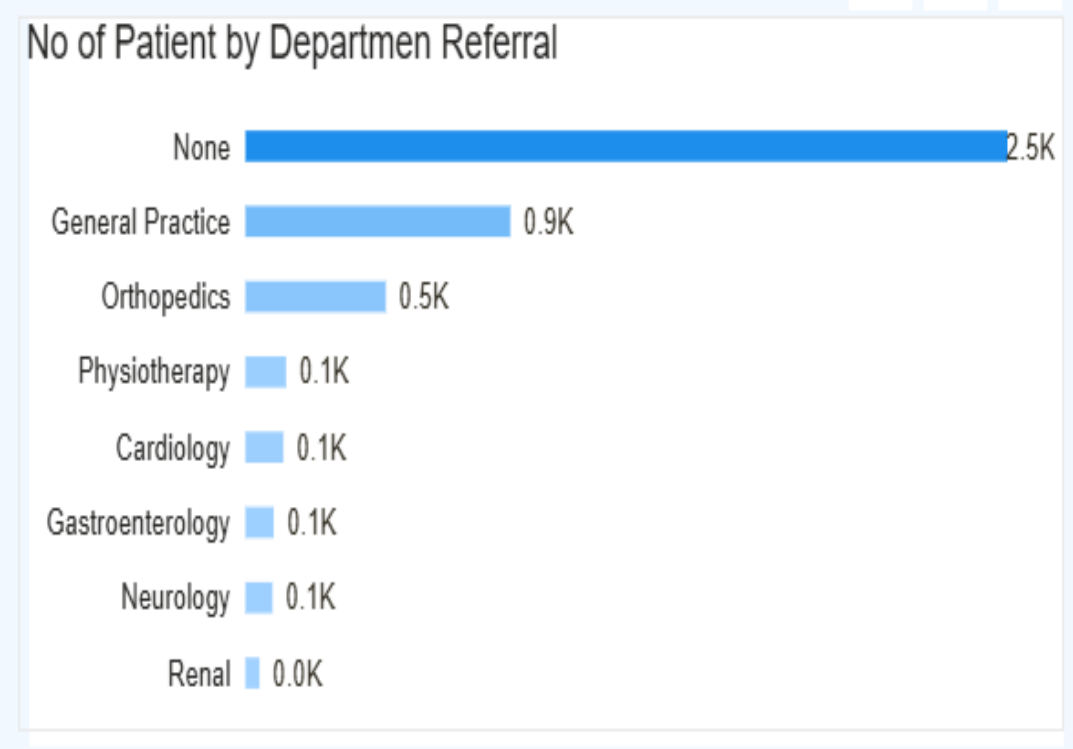
❖ Patient Gender Distribution

Patient visits are almost equally distributed between genders:
Male patients represent 50.67%
Female patients represent 49.15%
0.18% are not confirmed
Gender distribution analysis helps understand **patient demographics and healthcare utilization patterns**.



❖ Department Referral Analysis

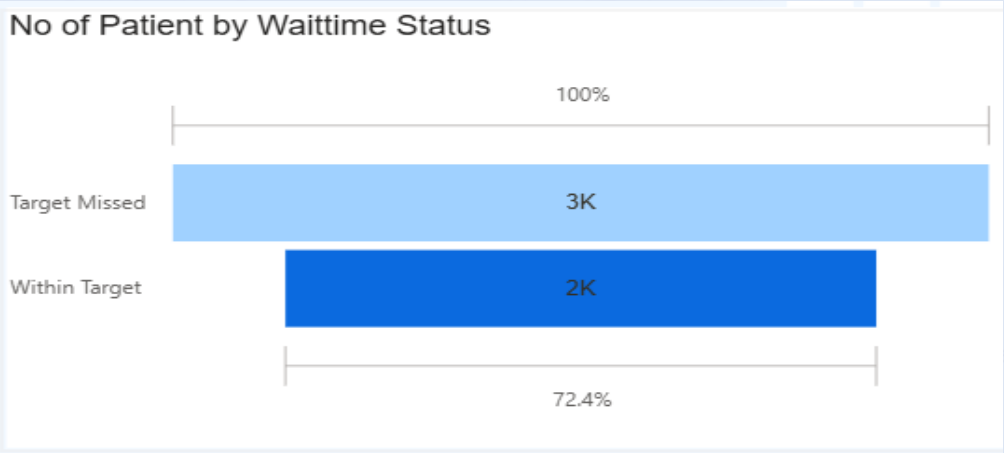
Most patients **did not require referral to other departments**, while some cases required specialized medical attention in departments such as:
General Practice
Orthopedics
Cardiology
Gastroenterology
Neurology
Referral patterns help hospitals analyze **specialized treatment demand**.



Dashboard 1: Monthly View

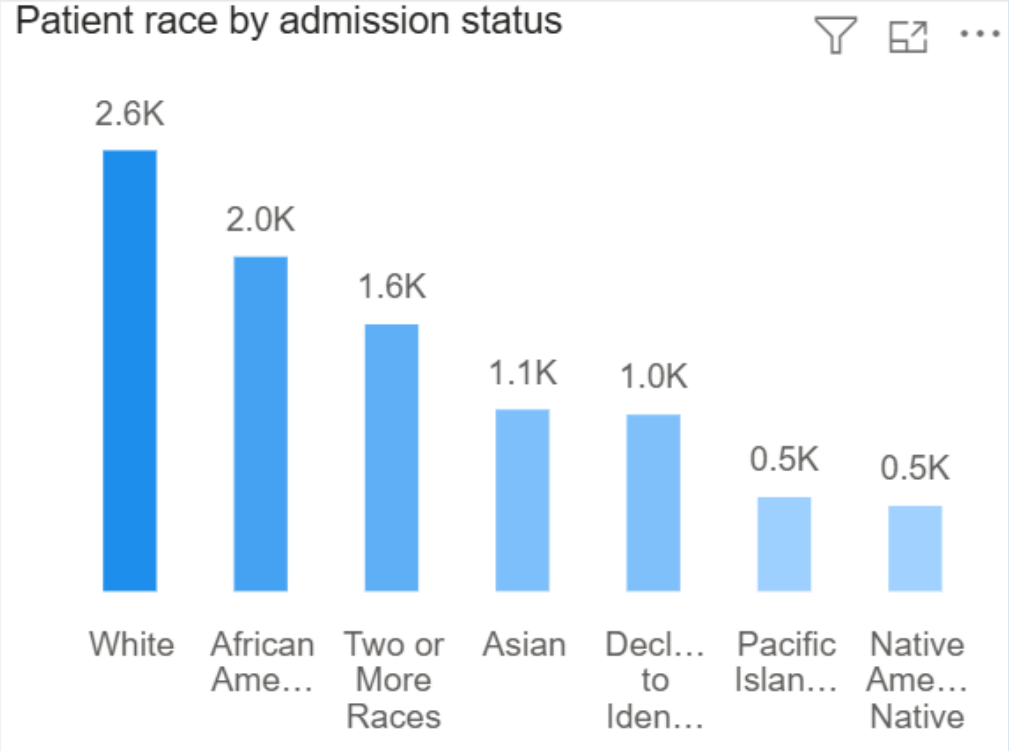
❖ No of Patient by Wait Time Status

Around **3K patients missed the target wait time**, while **2K patients were attended within 30 minutes**, indicating that most patients experienced **longer waiting times than the target**.



❖ Patient Race by Admission Status (ToolTip)

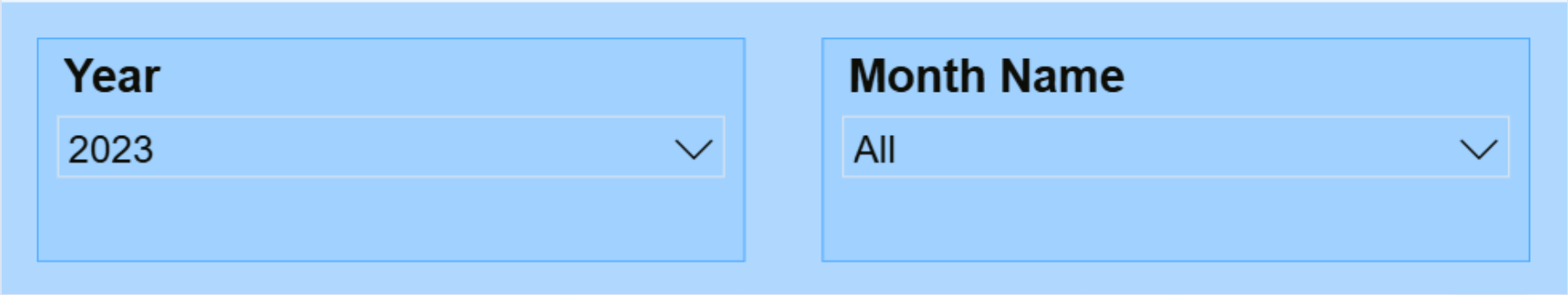
The majority of patients belong to the **White category (about 2.6K)**, followed by **African American (2.0K)** and **Two or More Races (1.6K)**, while **Pacific Islander and Native American groups have the lowest patient counts (about 0.5K each)**.



Dashboard 1: Monthly View

❖ **Slicers (Year & Month):**

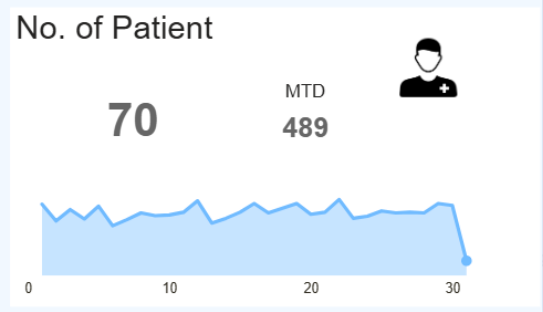
The **Year and Month slicers** allow users to filter the dashboard and analyze hospital emergency data for a **specific year or month**, helping to understand patient trends and performance over different time periods.



Dashboard 1: Monthly View

❖ **Drill Through Using KPI Card**

"No. of Patients" KPI card to the "Patient Admission Records" page for detailed analysis



PATIENTS ADMISSION RECORDS

Patient Name	Patient Id	Avg Wait Time	Patient Admission Date	Department Referral	Patient Gender	No of Patient
A Abarough	807-47-2097	Min	14 July 2023	None	Feamle	
A Addey	118-92-0340	55.0 Min	21 July 2024	None	Male	1
A Adney	721-88-0520	28.0 Min	27 May 2024	None	Feamle	1
A Albert	700-76-1641	40.0 Min	20 September 2024	None	Male	1
A Alejandro	399-73-4231	Min	25 July 2023	Gastroenterology	Feamle	
A Aleksich	372-20-5389	58.0 Min	13 July 2024	None	Feamle	1
A Algar	139-05-0602	Min	30 December 2023	None	Male	
A Alstead	355-73-0319	17.0 Min	21 May 2024	None	Feamle	1
A Alyukin	345-87-0533	Min	25 July 2023	None	Male	
A Ambrogioni	602-05-7479	Min	06 October 2023	Gastroenterology	Male	
A Angell	119-26-2201	27.0 Min	09 May 2024	None	Feamle	1
A Arden	302-70-1794	19.0 Min	24 June 2024	General Practice	Feamle	1
A Aspling	599-40-2753	20.0 Min	03 October 2024	None	Feamle	1
A Assad	827-40-8129	40.0 Min	03 January 2024	None	Male	1
A Attril	687-66-3054	Min	04 November 2023	None	Male	
A Backshaw	307-83-2857	21.0 Min	02 August 2024	None	Feamle	1
A Bagnal	475-34-5223	20.0 Min	12 July 2024	None	Male	1
A Barmadier	156-95-7483	Min	06 September 2023	General Practice	Feamle	
A Basili	413-63-3325	37.0 Min	23 August 2024	None	Male	1
A Batte	493-76-6065	13.0 Min	08 April 2024	None	Feamle	1
A Baudichon	739-19-5155	Min	13 October 2023	General Practice	Feamle	
A Beadnall	842-21-2088	Min	07 August 2023	Orthopedics	Male	
Total		**35.5 Min**				**4878**

Dashboard 2: Patient Details

Objective: **Offer granular insights into patient-level data to enable detailed analysis and troubleshooting.**

Charts to Develop: A grid displaying essential fields:

- Patient ID
- Patient Name
- Gender
- Age
- Wait Time
- Department Referral
- Admission Status
- Patient Race

BOOKMARKS

MONTHLY VIEW

PATIENT DEATILS

HOSPITAL EMMERGENCY DASHBOARD
MONTHLY VIEW
Apr 2023
Year
2023
Month Name
All
No. of Patient
4338
MTD
489
Avg Wait Time
35.0 Min
Patient Satisfaction Score
5.03
No. of Patient Reffered
1824
MTD
211
MONTHLY VIEW
PATIENT DEATILS
Patients Admission Status
Admission Status Patients % of Total
Admitted 2157
Not Admitted 2181
% of Patient Seen Within 30 Mins
No of Patient by Patient Race
White 1225
African American 920
Two or More Races 722
Asian 496
Declined to Identify 476
Pacific Islander 263
No of Patient by Age Group
543
546
527
536
578
556
570
482
70-79
60-69
50-59
40-49
30-39
20-29
10-19
0-9
No of Patient by Waittime Status
Target Missed
Within Target
No of Patient by Admission Status
2.2K
4.3K
Admitted
Not Admitted
Total
No of Patient by Departmen Referral
None 2.5K
General Practice 0.9K
Orthopedics 0.5K
Physiotherapy 0.1K
Cardiology 0.1K
Gastroenterology 0.1K
Neurology 0.1K
Renal 0.0K
No of Patient by Patient Gender
NotConfirmed 0.18%
Female 49.15%
Male 50.67%

Thank You